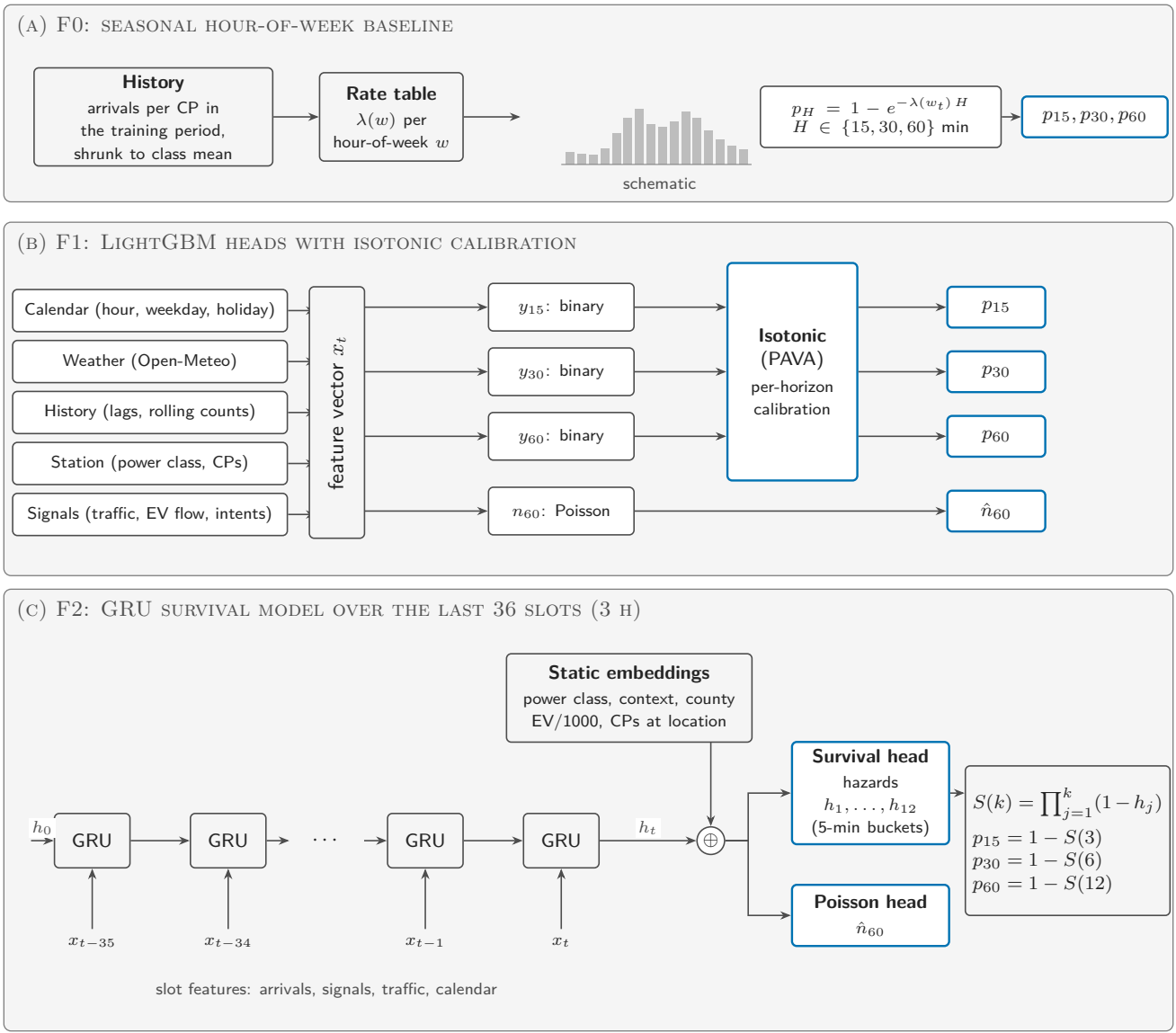

(A) F0: SEASONAL HOUR-OF-WEEK BASELINE
History
arrivals per CP in the training period, shrunk to class mean
Rate table
$\lambda(w)$ per hour-of-week w
schematic
$p_H = 1 - e^{-\lambda(w_t)\,H}$
$H \in \{15, 30, 60\}$ min
p_{15}, p_{30}, p_{60}
(B) F1: LIGHTGBM HEADS WITH ISOTONIC CALIBRATION
Calendar (hour, weekday, holiday)
Weather (Open-Meteo)
History (lags, rolling counts)
Station (power class, CPs)
Signals (traffic, EV flow, intents)
feature vector x_t
y_{15}: binary
y_{30}: binary
y_{60}: binary
n_{60}: Poisson
Isotonic (PAVA) per-horizon calibration
p_{15}
p_{30}
p_{60}
$\hat{n}_{60}$
(C) F2: GRU SURVIVAL MODEL OVER THE LAST 36 SLOTS (3 H)
Static embeddings
power class, context, county
EV/1000, CPs at location
h_0
GRU
GRU
· · ·
GRU
GRU
h_t
x_{t-35}
x_{t-34}
x_{t-1}
x_t
Survival head
hazards $h_1, \ldots, h_{12}$
(5-min buckets)
Poisson head
$\hat{n}_{60}$
$S(k) = \prod_{j=1}^{k}(1-h_j)$
$p_{15} = 1 - S(3)$
$p_{30} = 1 - S(6)$
$p_{60} = 1 - S(12)$
slot features: arrivals, signals, traffic, calendar